Ultrapowers

If (X,d) is a metric space, we may consider the space $\ell^\infty(X)$ to be the space of all bounded sequences in X with metric defined by

$$d_\infty(x,y) = \sup_{n\in\mathbb{N}} d(x_n,y_n).$$

If ω is an ultrafilter on $\mathbb{N}$, then we define an equivalence relation $x \sim y$ on $\ell^\infty(X)$ by saying they are equal if

$$\lim_\omega d(x_n,y_n) = 0.$$

Then, we define the equivalence classes

$$[x]_\omega = \{y \in \ell^\infty(X) \mid y \sim x\}$$

and take the quotient $X_\omega = \ell^\infty(X)/\sim$, with distance metric defined by

$$d_\omega\Big([x]_\omega,[y]_\omega\Big) = \lim_\omega d(x_n,y_n).$$

This space is known as the *ultrapower* of the metric space (X,d). The map $\iota_\omega\colon X \to X_\omega$ defined by $x \mapsto [(x,x,\dots)]_\omega$ is an isometric embedding of (X,d) into (X_ω,d_ω).

Generally, we assume that the ultrafilter ω is nonprincipal, since if ω is principal, then taking the limit along the ultrafilter is equivalent to extracting the element of $\mathbb{N}$ that defines the principal ultrafilter.

A useful fact is that if (X,d) is complete, then so too is (X_ω,d_ω).

Now, suppose G is a group, and let $d\colon G\times G \to [0,\infty)$ be a metric on G. We say the metric d is left (or right) invariant if $d(hg_1,hg_2) = d(g_1,g_2)$ for left invariance and $d(g_1h,g_2h) = d(g_1,g_2)$ for right invariance. A metric that is both left and right invariant is called bi-invariant, and the space (G,d) is called a metric group.

> **Example:** Let G be a finitely generated group, and let $S \subseteq G$ be a finite generating subset. Then, (G,d_S) is a left metric group, where d_S denotes the word metric — i.e., $d_S(g,h)$ is the minimum word length necessary to construct the element gh^{-1} from S.

In the next section, we will discuss two important metric groups and various important properties thereof.

Sofic Groups

Suppose S_n is the group of permutations of the set $[n] = \{0,\dots,n-1\}$. Then, the *Hamming length* of a permutation $\sigma \in S_n$ is given by

$$\ell(\sigma) = \frac{1}{n}|\{i\in[n] : \sigma(i) \neq i\}|.$$

The metric $d\colon S_n\times S_n \to [0,\infty)$ given by $d(x,y) = \ell\Big(xy^{-1}\Big)$ then is a bi-invariant length function.[I]

> **Definition:** A countable discrete group Γ is *sofic* if for every $\varepsilon > 0$ and every finite subset F of $\Gamma \setminus \{e\}$, there is n and a map $\Phi\colon \Gamma \to S_n$ such that $\Phi(e) = \mathrm{id}$, and for every $g,h \in F\setminus\{e\}$, we have
>
> - $d(\Phi(gh),\Phi(g)\Phi(h)) < \varepsilon$;
> - $\ell(\Phi(g)) > r(g)$, where $r(g)$ is a constant depending only on g.

[I] For invariance, recall that conjugation preserves cycle type, which is essentially what the Hamming distance is measuring. One can show that a metric being bi-invariant is equivalent to it being preserved under conjugation.

One of the main benefits of the ultrapower method is that it allows for a conversion from this local definition to a global definition. Fixing a free ultrafilter ω on $\mathbb{N}$, we define the ultraproduct

$$\prod_{\omega} S_n := \prod_{n\in\mathbb{N}} S_n \Big/ \left\{ x \in \prod_{n\in\mathbb{N}} S_n : \lim_{\omega} \ell_{S_n}(\sigma) = 0 \right\}.$$

It follows from this definition that Γ is sofic if and only if there exists an injective homomorphism $\Phi\colon \Gamma \to \prod_{\omega} S_n$ for any free ultrafilter ω on $\mathbb{N}$.

Definition: Suppose G and H are groups with invariant length functions, F a subset of G, and ε is a positive number. A function $\Phi\colon G \to H$ is called an (F,ε)-approximate homomorphism if $\Phi(e_G) = e_H$ and for every $g,h \in F$,

- $\left|\ell_H\left(\Phi(gh)\Phi(h)^{-1}\Phi(g)^{-1}\right)\right| < \varepsilon$;
- $|\ell_H(\Phi(g)) - \ell_G(g)| < \varepsilon$.

Theorem: A countable discrete group Γ is sofic if and only if for every $\varepsilon > 0$ and every finite $F \subseteq \Gamma$, there is n and an (F,ε)-approximate homomorphism from Γ to S_n, where Γ is regarded as an invariant length group with respect to the trivial invariant length function.

Here, the trivial invariant length function refers to the function

$$\ell_0(x) = \begin{cases} 0 & x = e_\Gamma \\ 1 & \text{else.} \end{cases}$$

Hyperlinear Groups

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